Classical Statistics

Describing dental data, and comparing two groups

HISTORY

Four tests before the computer, one after

1900

Chi-square test

Karl Pearson

mathematician, London

Counts in a table vs. counts expected if nothing is going on.

1908

t-test

W. S. Gosset – "Student"

brewer at Guinness, Dublin

Small samples of barley and beer.

1935

Fisher's exact

R. A. Fisher

statistician, Rothamsted

Small 2 × 2 tables where chi-square fails.

1945

Wilcoxon

Frank Wilcoxon

chemist at American Cyanamid

Ranks instead of values: quick by hand, no normality. Bug sprays

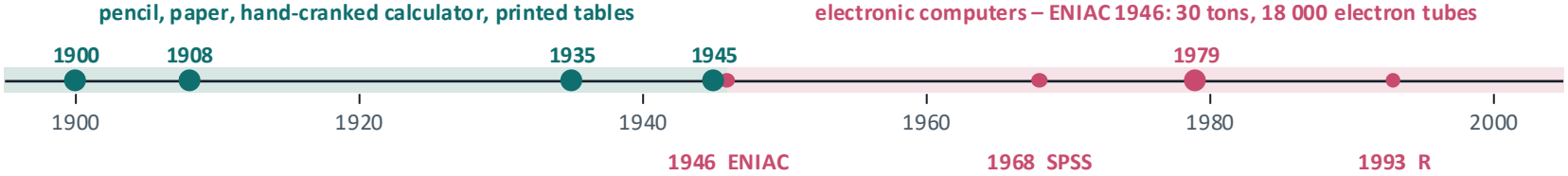
1979

Bootstrap

Bradley Efron

statistician, Stanford

Re-draw your own sample thousands of times: a CI for anything, no formula. Permutation.



The classical tests were built for pencil and paper – hence critical-value tables, and ranks instead of values. Resampling is the one idea that had to wait for the machine: thousands of re-draws give a p-value or a CI for anything – a median, a ratio, clustered teeth – with no formula and no normality.

ENIAC 1946 and the phone in your pocket



US Army photo, Moore School, University of Pennsylvania, 1946



	ENIAC 1946	iPhone 18 Pro, 2026	factor
switches	18 000 tubes	A20 Pro, 2 nm: tens of billions of transistors	× 1 000 000
speed	5 000 additions / s	trillions of operations / s	× 1 000 000 000
weight	30 tons	211 g	÷ 140 000
power	150 kW – a village	≈ 5 W, one day on a battery	÷ 30 000
memory	20 numbers	256 GB to 2 TB	× 10^{11}
price	$ 487 000 (≈ $ 8 million today)	$ 1 199	÷ 7 000

The plan for this session

1 - The data

What kind of number you have, and how to describe it honestly.

2 - The logic

Null hypothesis, alpha and beta, and what a p-value really asks.

3 - The tests

Continuous, ordinal, counts, survival - and correlation.

- Every example is dental: probing depth, pain after third molar surgery, dentures in the over-seventies, implant survival.
- A little R along the way - enough to run each test

1

What statistics is for

Describing what you measured, and inferring what it means

IN GENERAL

Two jobs, not one

Descriptive statistics

Organise and summarise what you actually measured: the mean probing depth in your patients, the shape of the distribution, the spread.

Inferential statistics

Use that sample to say something about patients you did not measure: is protocol A really better, or did chance produce the gap?

The order matters

Description comes first.

IN GENERAL

Why a clinician needs this

42-78%

of original papers in critical reviews of selected medical journals used an inappropriate method of analysis

Gaddis & Gaddis, Ann Emerg Med 1990, Part 1

73%

of articles were fully accessible with descriptive statistics, t-tests and contingency tables - 58% on descriptive statistics alone

Emerson & Colditz, N Engl J Med 1983, 760 NEJM articles

- The analysis is often delegated - but the author remains responsible for the conclusions.
- You cannot plead ignorance when the method was wrong; you signed the paper. **Do univariate analyses yourself**
- **Most of what you need is in this session.**

IN GENERAL

Sample and population

Population

Everyone you want to speak about - every patient with periodontitis, not just yours. Usually impossible to measure.

Sample

The subset you actually study. **Conclusions travel back to the population only if the sample represents it (think selection bias).**

Random selection is the bridge

Every member of the population needs an equal and independent chance of being chosen. Where that is violated, bias enters - and no analysis removes it afterwards.

IN GENERAL

Parametric or non-parametric

Parametric methods

Assume the data follow a known distribution, normally the Gaussian curve. For interval and ratio data: probing depth, age, insertion torque.

We work with distributions

Non-parametric methods

Assume far less - usually only that values can be ranked. For ordinal data, and for small or clearly skewed samples.

We work with ranks

The scale decides

Ordinal data - plaque index 0 to 3, mobility grade I to III, a four-point pain grade - belong in the second column

2

Describing data

The four scales, the shape of the distribution, and how to summarise it

Four scales: N-O-I-R

Nominal

Labels, no order. Sex, tooth type, denture yes or no.

Ordinal

Ranked, unequal steps. Pla**que index 0 to 3, mobility grade I to III**

Interval

Equal steps, arbitrary zero. **Overjet (0 is edge to edge), Year of treatment, pH of plaque (pH 0 is not no acidity)**
Ratios make no sense

Ratio

Equal steps, real zero. Probing depth, age, cost, number of teeth, pain on a VAS.

Why this comes first

Interval and ratio data may be analysed with parametric methods; ordinal and nominal data may not. The scale fixes the test before you have seen a single number.

What each scale lets you do

	Mean	Median	Mode	Affected by outliers
Ratio - probing depth in mm	yes	yes	yes	the mean only
Ordinal - plaque index 0 to 3	no	yes	yes	no
Nominal - denture yes / no	no	no	yes	no

The mean only makes sense for interval and ratio data

The normal distribution

Sample SD

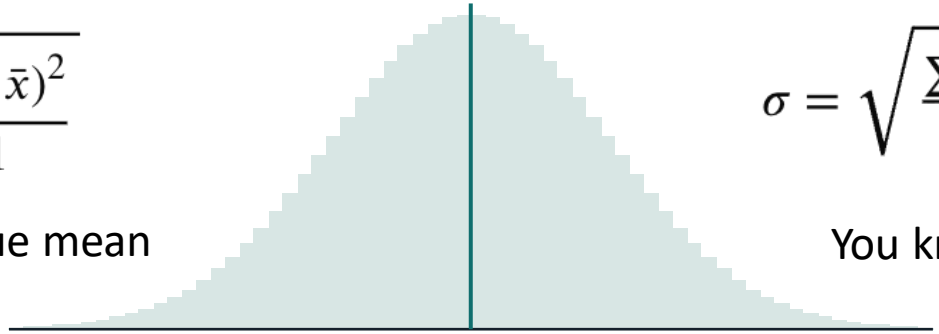
$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

You dont know the true mean
n-1 adjusts for that

Population SD

$$\sigma = \sqrt{\frac{\sum(x_i - \mu)^2}{n}}$$

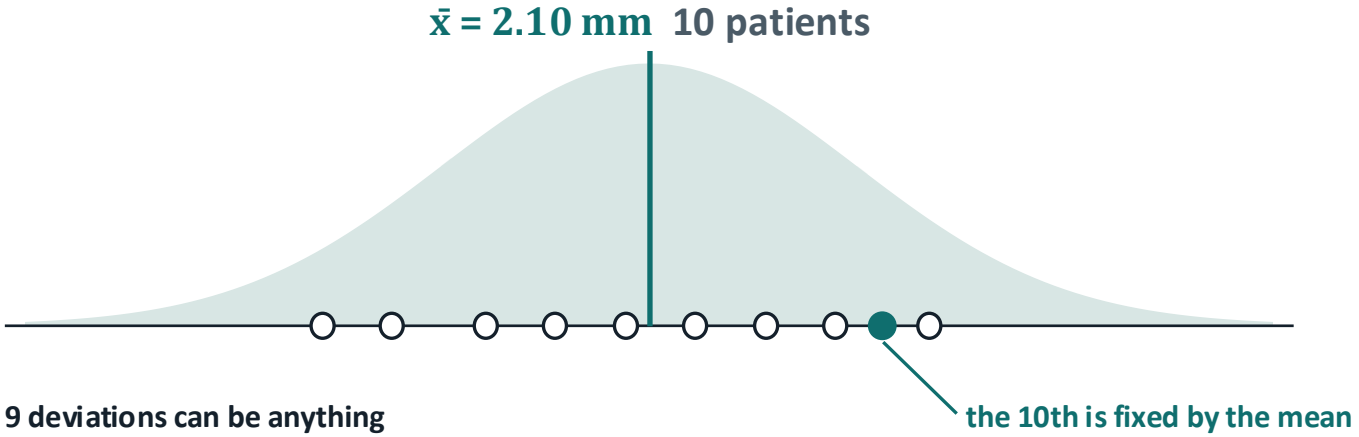
You know the true mean



mean = median = mode

- Symmetrical and bell shaped; described completely by two numbers, the mean and the SD.
- **Near enough normal in dentistry: mesiodistal crown width, marginal bone level, insertion torque of an implant (per bone density class)**
- **Not everything is: cost, waiting time and pocket depth at diseased sites all have a long right tail**.

Why n – 1: the tenth deviation cannot vary



Degrees of freedom: how many deviations are free to vary. Ten patients, one mean – nine deviations can take any value, **the tenth is whatever brings the mean to 2.10**.

Dividing by n − 1 is the price of guessing the population from a sample. If you have the whole population you are not guessing, so you pay nothing – divide by N.

and the shapes it is not

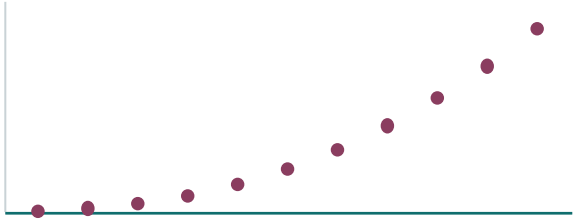
Bimodal | Rectangular | Positively skewed | Negatively skewed

- **Bimodal**: teeth remaining in the over-seventies - a cluster at zero, another near 25, and a mean of 12 that fits nobody.
- **Rectangular**: every value about equally often – randomisation (random value generator), month of birth of patients
- **Positively skewed: a long right tail** - treatment cost, waiting time,
- **Negatively skewed**: a long left tail - satisfaction scores, where most patients tick 9 or 10.
- **When the shape is not normal, the mean stops being a fair summary.**

Is it normal enough? Look at the shape



normal: the points follow the diagonal



skewed right: concave upward

- A Q-Q plot sets your ordered values against the values a normal distribution would have produced.
- Concave upward is a long right tail, an S-shape means heavy tails, and steps mean too few distinct values.
- **Draw the histogram and the box plot beside it**: the same information, more coarsely, and easier to read out loud.

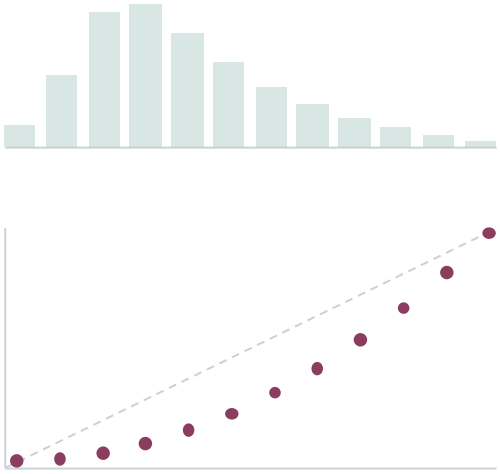
What the picture tells you

Which kind of departure you have - a long tail, a second peak, too few distinct values

The three departures, side by side

A long right tail

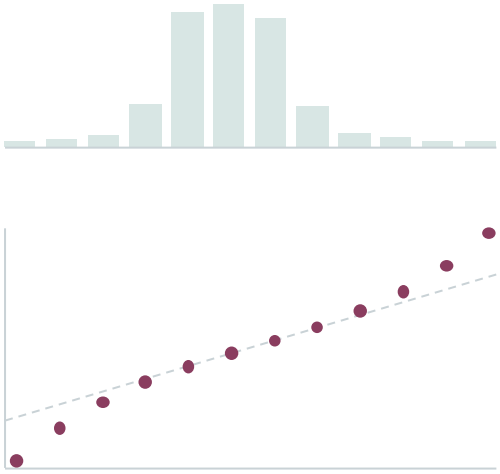
cost, waiting time, lesions per patient



Q-Q: concave upward

Heavy tails

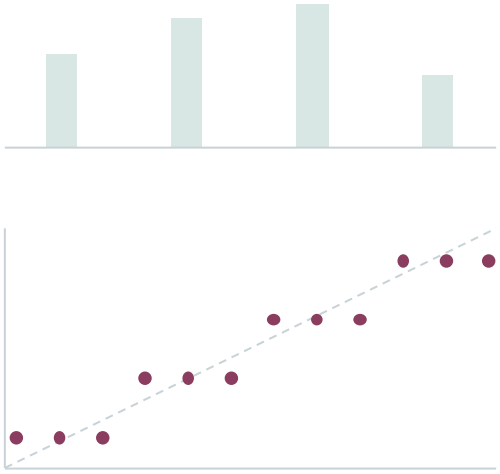
a few extreme patients at both ends



Q-Q: an S-shape

Too few distinct values

a graded index, 0 to 3



Q-Q: steps

The histogram shows the shape; the Q-Q plot shows how it leaves the diagonal - read them side by side.

Mean, median and mode

Mean

The arithmetic average. Interval and ratio data only. Pulled about by outliers - one 9 mm pocket moves it.

Median

The middle value, the 50th centile. Works for ordinal data and ignores outliers.

Mode

The most frequent value. The only one that means anything for nominal data such as denture yes or no.

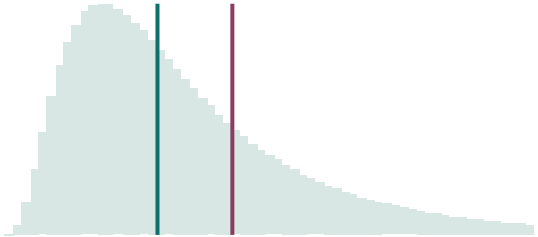
No single measure is best

In a normal distribution all three coincide

Where the three measures sit

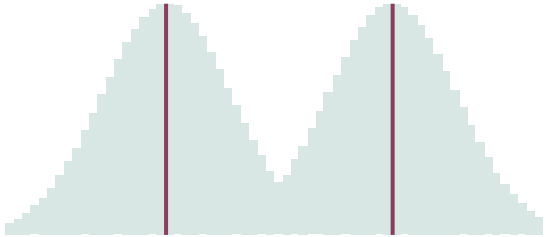
symmetrical: all three equal

crown width of an upper central incisor



skewed: median, then mean

cost of a course of treatment



bimodal: two modes, no useful mean

teeth remaining in patients over 70

Reading a paper

Mean and median far apart is the fastest signal that the data are skewed - and that a t-test may be the wrong tool.

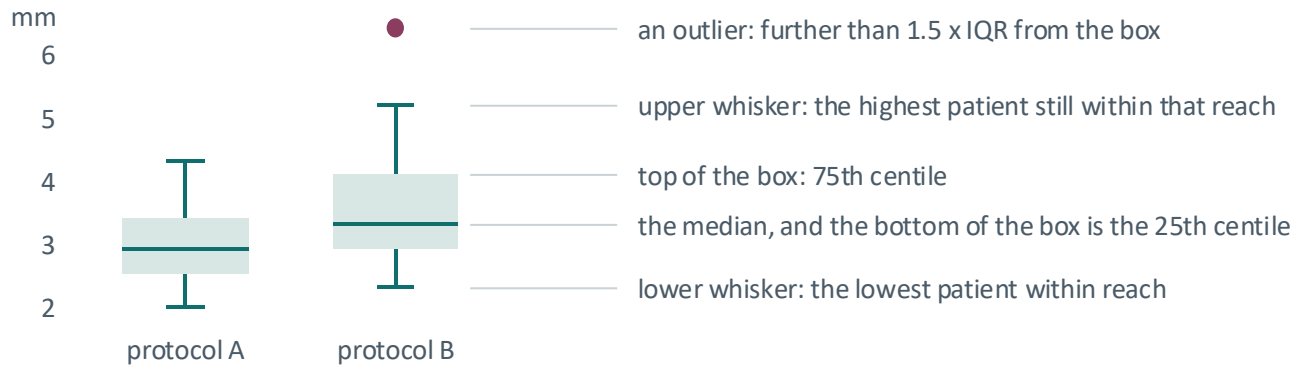
Four ways to report the spread

	What it is	Use it with	In R
Range	Lowest to highest	a first look, any scale	range(x)
Interquartile range	25th to 75th centile	ordinal or skewed data	IQR(x)
Standard deviation	Typical distance from the mean	normal-ish data	sd(x)
Standard error	SD / square root of n	the precision of the mean	sd(x)/sqrt(n)

Never a mean without a spread

A mean probing depth of 3.4 mm says nothing on its own. One outlier moves the range and the mean, and leaves the median untouched.

The box plot: five numbers in one picture

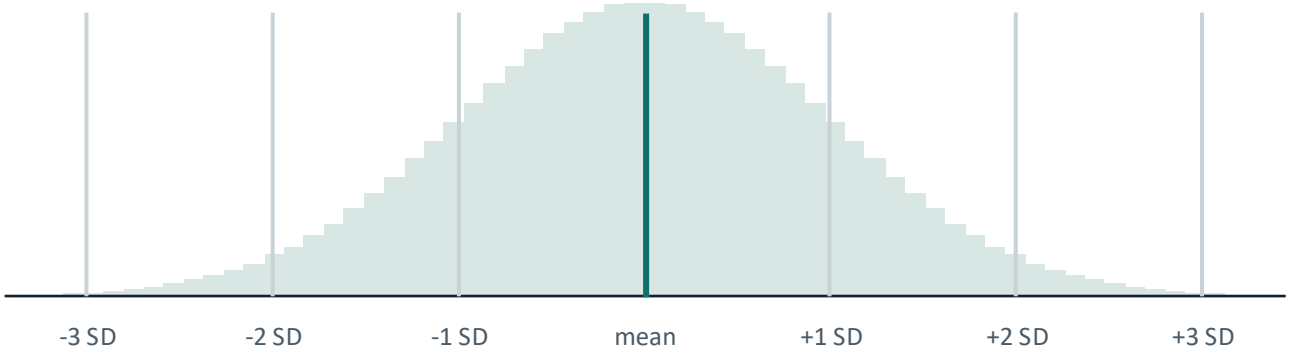


- The box holds the middle half of the patients, from the 25th to the 75th centile - the interquartile range, IQR.
- The whiskers stop at the most extreme patient that still lies **within 1.5 x IQR of the box**; anything past that is drawn as its own dot and deserves a look at the chart.

In R

boxplot(a, b) for two vectors, or boxplot(depth ~ group, data = d) for a data frame. Not a bar chart of means.

The standard deviation and the normal curve



68.27%

of patients within 1 SD of the mean

95.45%

within 2 SD

99.73%

within 3 SD

Standard deviation or standard error?

SD describes the patients

How far individual patients scatter around the sample mean. It does not shrink when you recruit more of them.

SEM describes the mean

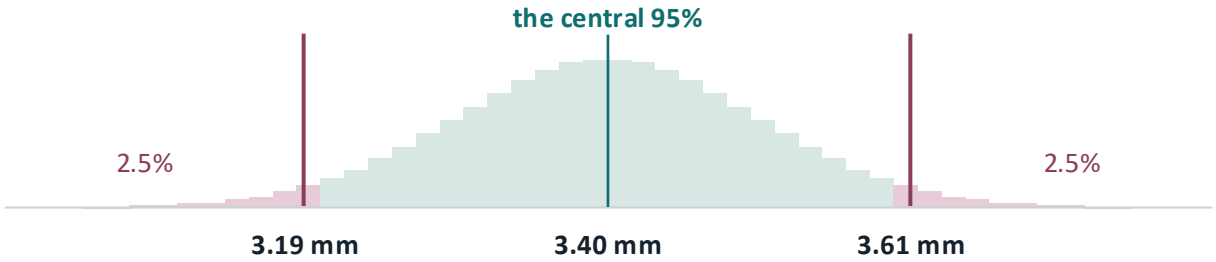
How far sample means would scatter around the true value. SEM = SD / sqrt(n)

so it shrinks with every patient -> higher sample size -> higher precision

Reported the wrong way round more often than you would think

Nagele reviewed 860 papers in four anaesthesia journals: one in four gave the SEM where only the SD was appropriate; in 441 cardiovascular papers of 2012 the figure was 64%. The SEM is simply smaller, and makes the data look tidier than they are. **Multiply it by the square root of n to get the SD back**.

The 95% interval, and where 1.96 comes from

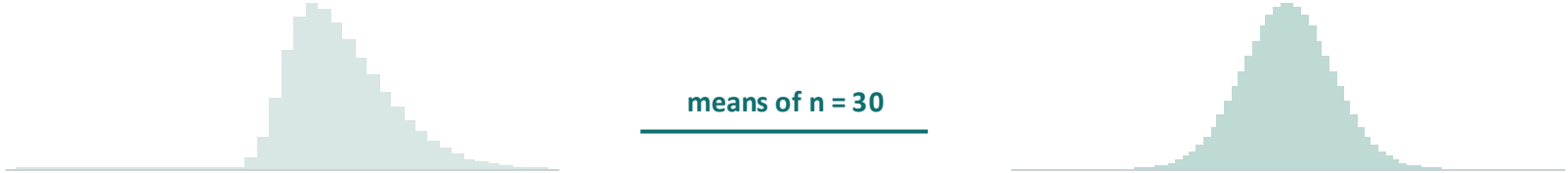


95% CI = mean +/- 1.96 x SEM = 3.40 +/- 1.96 x 0.11 = 3.19 to 3.61 mm

30 patients, mean probing depth 3.40 mm, SD 0.60 mm -> SEM = SD / sqrt(n) = 0.60 / sqrt(30) = 0.11 mm

- 1.96 is not magic: 95% of any normal curve lies within 1.96 standard deviations of its centre, leaving 2.5% in each tail.
- **The curve drawn here is not the patients but the means**. The SD of 0.60 mm describes how the 30 patients scatter; the SEM of 0.11 mm says how well the average is pinned down.
- **In a small study replace 1.96 by the slightly larger t value (2.05 at n = 30). R does that for you.**

Why the normal curve keeps coming back



one sample of patients, skewed

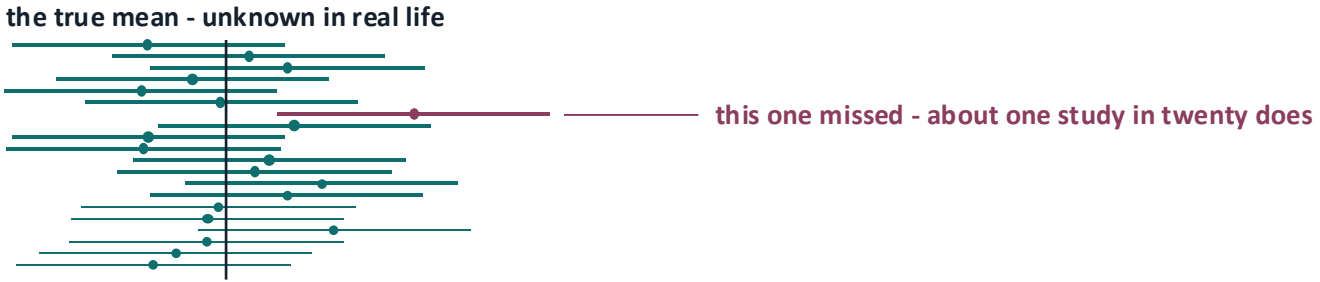
the means of many such samples

- Draw samples again and again and write down each mean: those means scatter normally around the true value, whatever shape the raw data had. **This is called the central limit theorem**.
- **The more patients, the closer to normal. With ten patients and skewed data the means are still skewed — the theorem has not kicked in yet. Rule of thumb: 30 patients; the heavier the skew, the more you need. You can check this yourself with a bootstrap (next lesson)**

What the theorem buys you

It is why the t-test and the confidence interval still work on data that are not perfectly normal

What '95% confident' actually means



twenty studies, twenty different intervals

- **Repeat the study and every sample gives a slightly different interval. In the long run 95 out of 100 of them contain the true value.**
- **So the 95% belongs to the method, not to your one interval: yours either covers the truth or it does not, and you cannot tell which.**

3

The logic of a test

Null hypothesis, alpha and beta, and the two ways of being wrong

THE LOGIC

The null hypothesis

H0

Protocol A and protocol B come from the same population: the mean probing depth after treatment is the same.

- H1, the alternative, says a difference really exists.
- The test examines H0 - you reject it, or you fail to reject it.
- You never prove H0. 'Not significant' means 'not enough evidence', not 'no difference'.

THE LOGIC

What a p-value actually asks

The question behind every test

If the two protocols really worked equally well, how often would chance alone hand me a gap at least as large as the one I am looking at?

- **p = 0.23 means: results like these turn up in about a quarter of studies when nothing is going on.**
- **p = 0.01 means: results like these turn up in about 1 in 100 studies when nothing is going on.**
- **Any cut-off is arbitrary — 0.05 is convention, 0.01 or 0.005 are defensible. But it is chosen before the data are in.**
- It is not the probability that H0 is true.
- It is not the size of the effect, and not the chance the result will replicate.

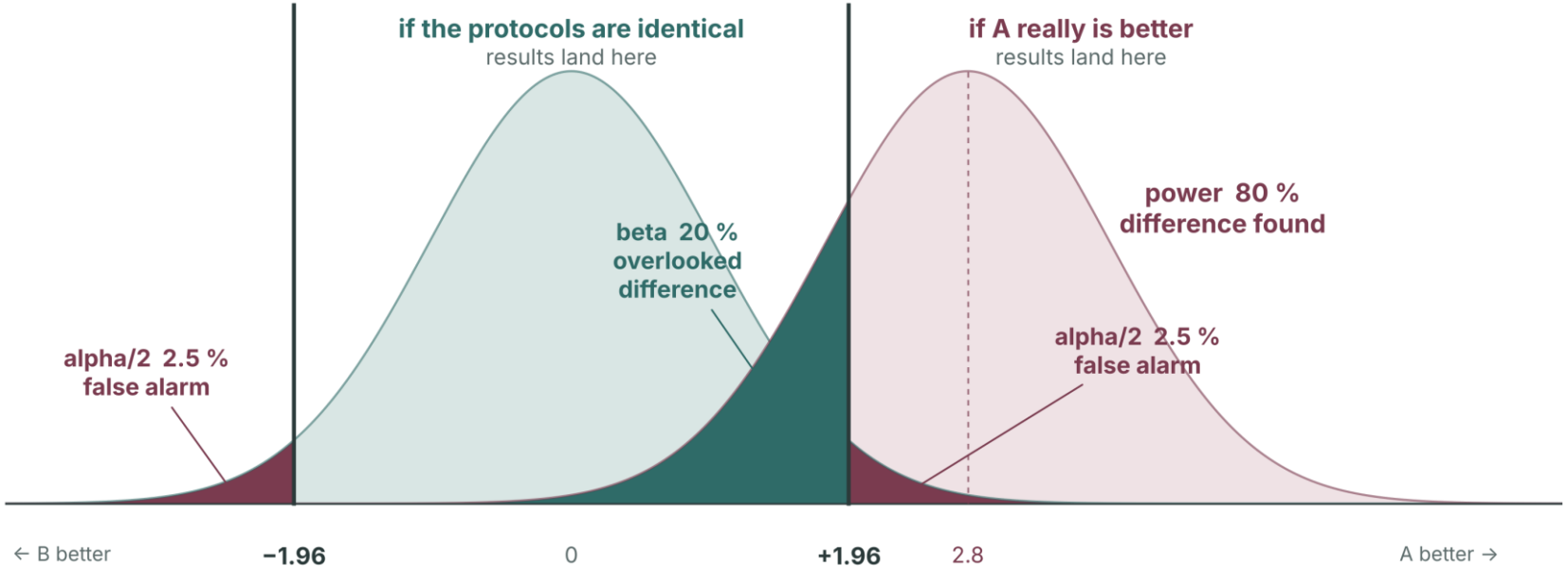
Two ways of being wrong

	The protocols really are equal	They really differ
You claim a difference	Type I error - false positive (alpha)	Correct - true positive (**power, 1 - beta**)
You claim no difference	Correct - true negative (1 - alpha)	Type II error - false negative (beta)

In one sentence

Alpha is the risk of announcing a difference that is not there. Beta is the risk of overlooking one that is. Convention: alpha 0.05, beta 0.20 (power 0.80). Power is bought with patients before the study — not discovered afterwards.

Alpha, beta and power: two-sided



(mean A − mean B) / standard error of the difference

How many standard errors apart are the two groups?

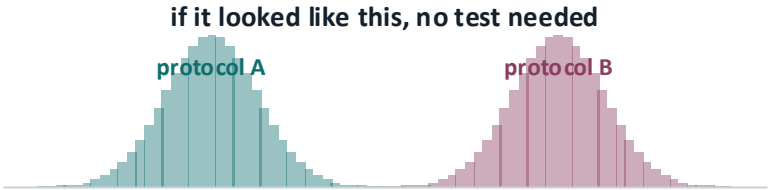
4

Comparing two means

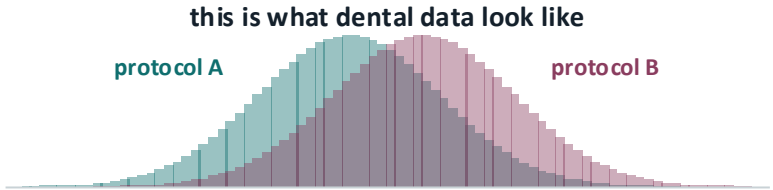
Probing depth after two protocols: Student's t-test

What is a t-test?

It asks one question: do the two averages differ by more than chance would produce?



the two groups barely share a patient



the patients overlap almost completely

each curve is the spread of individual patients, in millimetres of probing depth reduction

The test is not about the patients

Overlap on the right is the normal case, and it is why the question needs a test at all. The t-test does not compare patients: it compares the two averages, and an average is far less variable than the patients behind it.

WHAT IT IS

Three t-tests, three situations

One sample

One group against a number that is already known.
Your 30 patients against the 1.7 mm reported in the literature.

Two independent samples

Two separate groups of patients.
Ten treated with protocol A, ten with protocol B.

Paired samples

One group, measured twice.
The same mouth: left side treated with A, right side with B.

Split mouth design

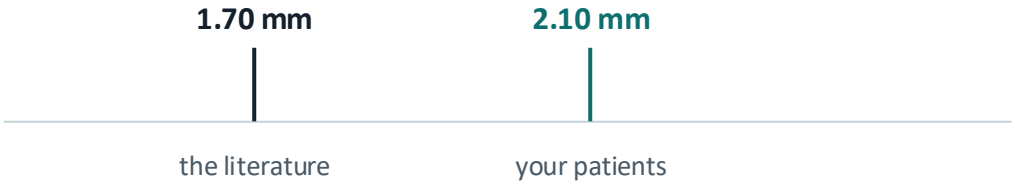
Which one you need is decided by the design, not by the data

Ask first: how many groups of patients are there, and is every measurement from a different mouth?

TYPE 1

One sample: against a known value

The literature says that standard therapy reduces probing depth by 1.7 mm.
You treat 30 patients your own way and measure a mean reduction of 2.10 mm.



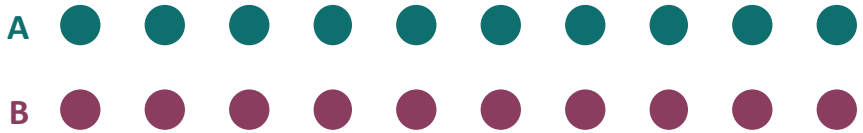
The question

Is a gap of 0.40 mm enough to say that your result really differs from the published value - or is 30 patients simply too few to tell?

TYPE 2

Two independent samples

Twenty patients, split at random: ten receive protocol A, ten receive protocol B.
Nobody appears in both groups, and no patient is measured twice.



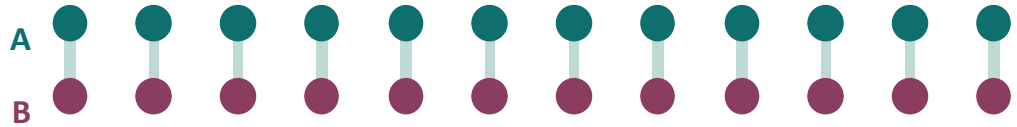
The question

Protocol A reduced pockets by 2.22 mm and protocol B by 1.72 mm. Is that gap of 0.50 mm more than chance would hand you?

TYPE 3

Paired samples (eg split mouth design)

Twelve patients, split mouth: the left side is treated with A, the right side with B. Every patient supplies both numbers, so the two measurements belong together.



one patient, one pair of measurements

The question

Within each mouth, side A did better by 0.5 mm on average. Is that difference real, or would chance produce it in twelve patients?

A USEFUL TRICK

The paired test is the one-sample test in disguise

Patient	Side A	Side B	Difference	
1	2.4	1.8	0.6	
2	3.1	2.2	0.9	
3	2.0	1.6	0.4	and so on

Once you have the last column, the two sides never appear again

You are left with one column of differences, and a reference value of zero - which is exactly a one-sample t-test. That is why the paired test needs so many fewer patients: the differences between patients have already been subtracted away.

THE ARITHMETIC

The same division, three times

	Difference	Standard error	df
One sample	your mean minus the known value	s / sqrt(n)	n - 1
Two samples	mean A minus mean B	sp x sqrt(1/nA + 1/nB)	nA + nB - 2
Paired	the mean of the differences	s of the differences / sqrt(n)	n - 1

Read the middle column as one idea

The standard error always says the same thing: how much the mean would move if you repeated the study with new patients. In the middle row sp is the pooled SD - with equal group sizes simply $sp^2 = (sA^2 + sB^2)/2$. sp^2 is the pooled variance

WORKED

One sample: 30 patients against 1.70 mm

$$t = (2.10 - 1.70) / (0.90 / \mathrm{sqrt}(30)) = 0.40 / 0.164 = 2.44$$

mean 2.10 mm, SD 0.90 mm, n = 30, reference value 1.70 mm

- The gap is 0.40 mm; the uncertainty of your mean is 0.164 mm; so the gap is 2.44 uncertainties wide.
- Degrees of freedom: 30 - 1 = 29. The mean costs 1 df
- **The table's critical value at 5% and 29 df: 2.045 - and 2.44 is beyond it.**
- That is far enough out to matter: p = 0.021.

What you would write

Mean reduction 2.10 mm (SD 0.90), which is 0.40 mm more than the published 1.70 mm; p = 0.021.

The t table

two-sided, the usual case

df	alpha 0.10	alpha 0.05	alpha 0.01
1	6.314	12.706	63.657
2	2.920	4.303	9.925
5	2.015	2.571	4.032
9	1.833	2.262	3.250
11	**1.796**	**2.201**	**3.106**
15	1.753	2.131	2.947
18	**1.734**	**2.101**	**2.878**
29	**1.699**	**2.045**	**2.756**
60	1.671	2.000	2.660
very large	1.645	1.960	2.576

How to use it

Pick your alpha, find your degrees of freedom, read the number - and reject the null hypothesis if your own t is larger. The three rows in teal are the examples from the last slide.

WORKED

Two samples: protocol A against protocol B

$$t = (2.22 - 1.72) / \text{sqrt}(0.805 \times (1/10 + 1/10)) = 0.50 / 0.40 = 1.25$$

group A: mean 2.22 mm, SD 0.79 mm, n = 10 · group B: mean 1.72 mm, SD 0.99 mm, n = 10
pooled variance $sp^2 = (0.79^2 + 0.99^2)/2 = (0.62 + 0.99)/2 = 0.805$ - the variance / standard error of the previous slide

- The gap of 0.50 mm is only one and a quarter times its own uncertainty of 0.40 mm.
- Degrees of freedom: 10 + 10 - 2 = 18.
- **The table's critical value at 5% and 18 df: 2.101 - and 1.25 falls short.**
- p = 0.23 - chance produces gaps this large about once in every four studies of identical protocols.

What you would write

Mean reduction 2.22 mm with A and 1.72 mm with B, difference 0.50 mm (95% CI -0.34 to 1.34); p = 0.23.

The t table

two-sided, the usual case

df	alpha 0.10	alpha 0.05	alpha 0.01
1	6.314	12.706	63.657
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60	1.671	2.000	2.660
very large	1.645	1.960	2.576

How to use it

Pick your alpha, find your degrees of freedom, read the number - and reject the null hypothesis if your own t is larger. The three rows in teal are the examples from the last slide.

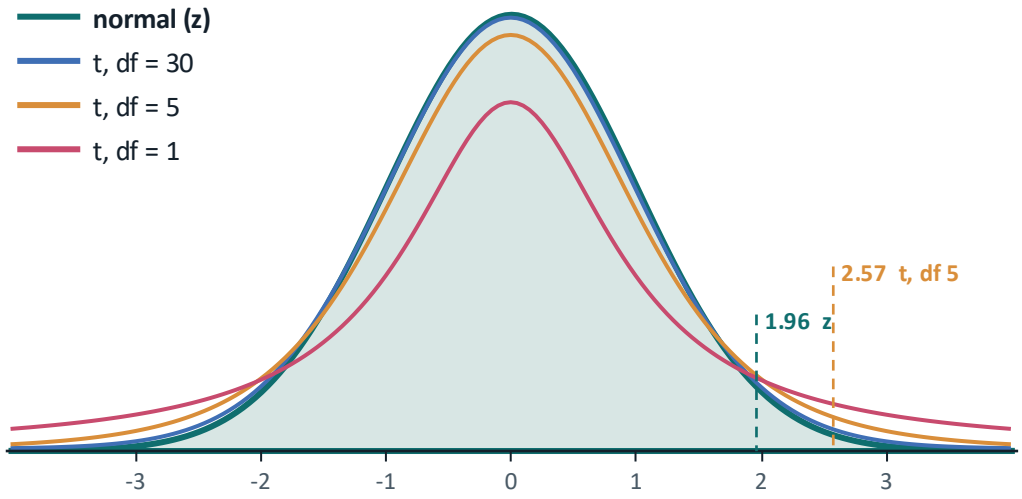
ROUTE 1

Reading a critical value from the table

- Choose alpha - say 0.05 - and note that two-sided means 0.025 in each tail.
- Count the degrees of freedom: 29 for the one-sample example, 18 for the two groups.
- Look the value up, and compare it with the t you calculated.

Example	df	Critical t at 5%	Your t
One sample, 30 patients	29	2.045	**2.44 -> reject H0**
Two groups of ten	18	2.101	1.25 -> keep H0

The t-distribution: a normal curve with heavier tails



Same shape, same centre. Bell, symmetric around 0.

Heavier tails, flatter peak. More probability far out – deliberately cautious.

.**One parameter: df = n – 1.** Few df → heavy tails. From about 30 df it is practically the normal.

Same difference, wider interval: with 6 patients the 95 % cut-off is 2.57 instead of 1.96.

WHAT NOW

You have a t. There are two ways to use it

Route 1: the critical value

Look up the value that t must beat in a table, for your alpha and your degrees of freedom.
The way it was done before computers, and still the way to understand it.

Route 2: the p-value

Let the software convert t directly into a probability.
Same decision, more information: you see how close the call was.

IN R

All three, in three lines

```
t.test(depth, mu = 1.7)                    # one sample against a known value
t.test(a, b, var.equal = TRUE)             # two independent groups
t.test(left, right, paired = TRUE)         # split mouth

qqnorm(depth); qqline(depth)               # look at the shape first
```

Two defaults worth knowing

Without var.equal = TRUE, R runs Welch's version, which does not assume equal variances - usually the better choice, but say which one you reported. And without paired = TRUE, a split-mouth study is analysed as though the sides came from different people.

Reading the output

Output	Value	What it means
t	1.25	The gap is 1.25 standard errors from zero
df	18	10 + 10 - 2
p-value	0.229	Chance alone would do this often
95% CI	-0.34 to 1.34 mm	Includes zero, so no difference is shown
means	2.22 and 1.72 mm	Always report these as well

5

How many patients?

How many patients a t-test needs, and what a confidence interval adds

What a sample size calculation needs

- The difference you would not want to miss - 0.5 mm, or 1 mm? **A clinical judgement**, not a calculation.
- **The spread of the outcome**, from a pilot study or from the literature.
- **Alpha**, the false-positive risk you accept: usually 0.05.
- **Power**, the chance of finding the difference if it is there: usually 0.8

CAVE

You should add patients for expected loss to follow-up, otherwise your study will be underpowered

The formula behind the numbers

$$n \text{ per group} = 2 \times (z(\text{alpha}/2) + z(\text{beta}))^2 \times SD^2 / \text{difference}^2$$

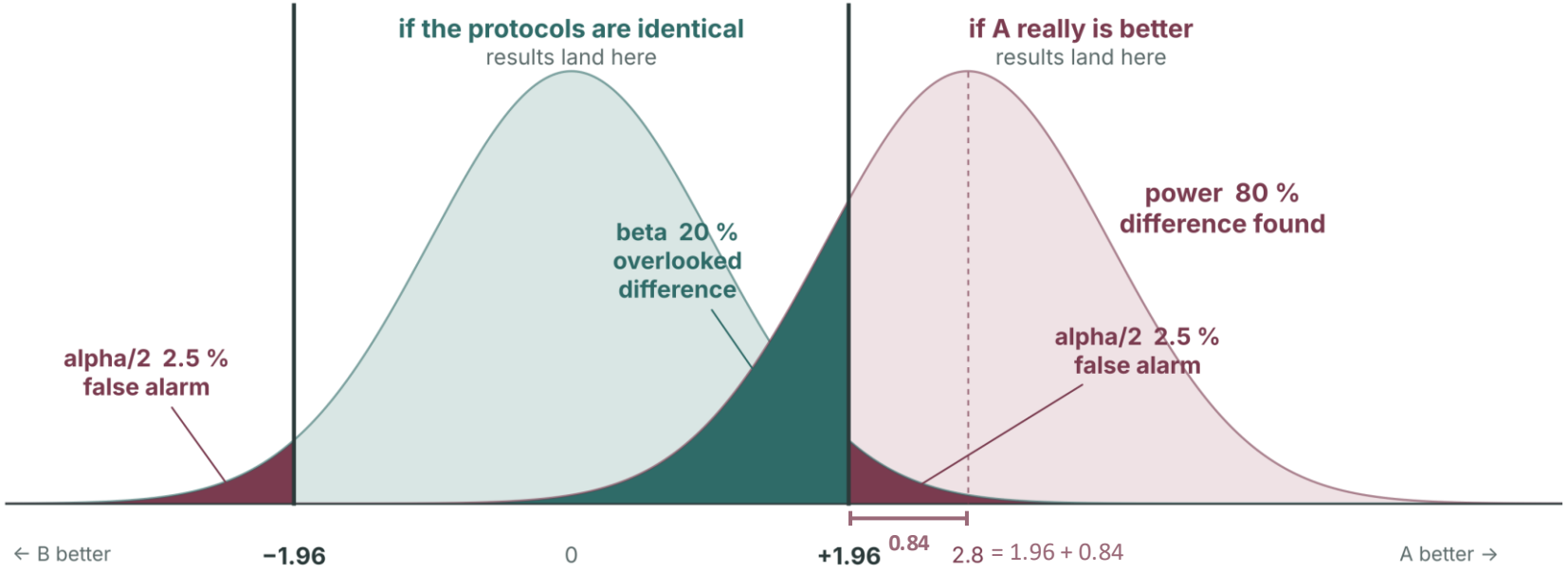
at alpha 0.05 two-sided and power 0.80 the two z values are 1.96 and 0.84, so the constant $2 \times (1.96 + 0.84)^2 = 15.7$

- Probing depth: SD 0.9 mm, and 0.5 mm is the difference worth finding. n = 15.7 x 0.81 / 0.25 = 51 patients per group.
- z, not t, because before the study you do not know n - and therefore not the degrees of freedom either. power.t.test uses t and returns a slightly larger number; **round up, then add for dropouts**.

Read the formula, not the software

The difference sits squared in the denominator: halving the difference you look for quadruples the study.

Alpha, beta and power: two-sided



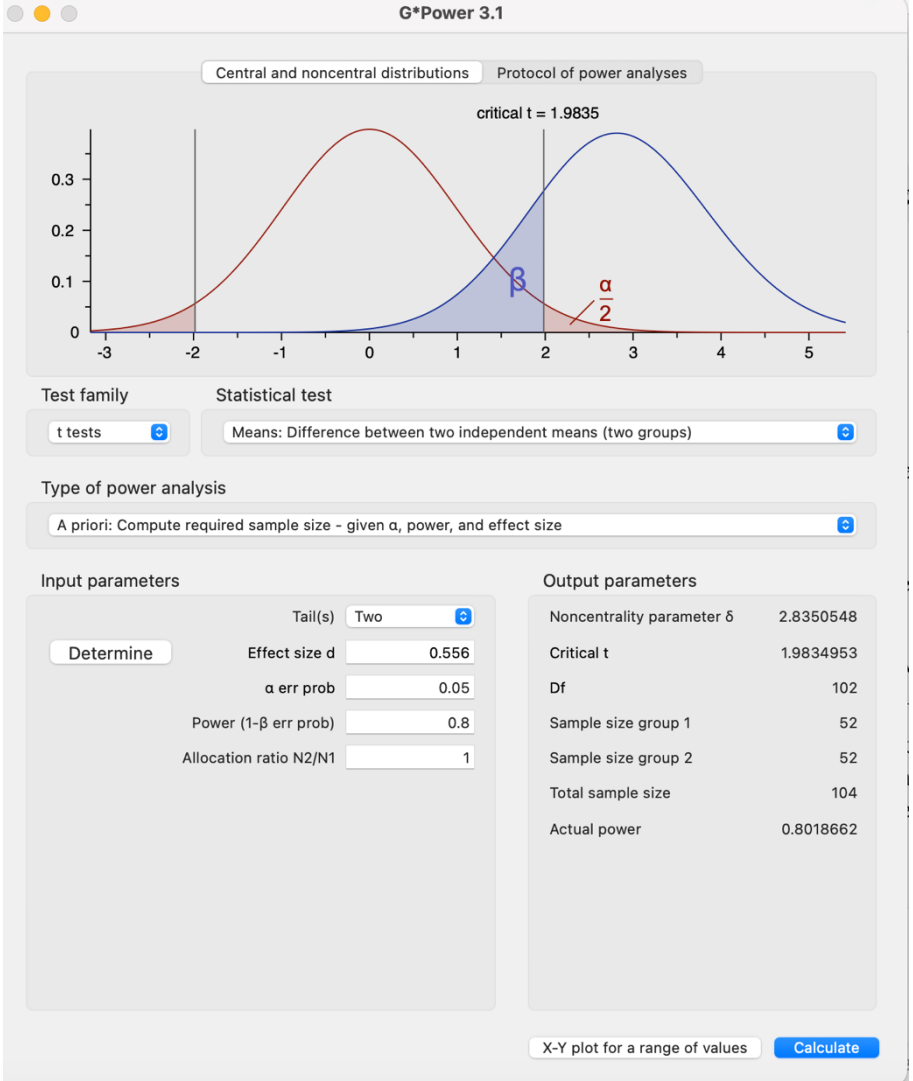
(mean A − mean B) / standard error of the difference

How many standard errors apart are the two groups?

Sample size in R

```
# from the pilot: difference 0.50 mm, SD about 0.90 mm
power.t.test(delta = 0.5, sd = 0.9, sig.level = 0.05, power = 0.80)

# or ask what 10 per group could have detected
power.t.test(n = 10, delta = 0.5, sd = 0.9)
```



SAMPLE SIZE

G*Power wants Cohen's d

Free download

Heinrich Heine University Düsseldorf, Windows and macOS: psychologie.hhu.de → G*Power.

Cohen's d = difference / SD

A unit-free effect size: the difference expressed in standard deviations. Here d = 0.5 mm / 0.9 mm = 0.56.

Why the numbers differed from R

power.t.test takes delta and sd separately, in mm. G*Power collapses both into d before it starts.

d = 0.5 is not 0.5 mm

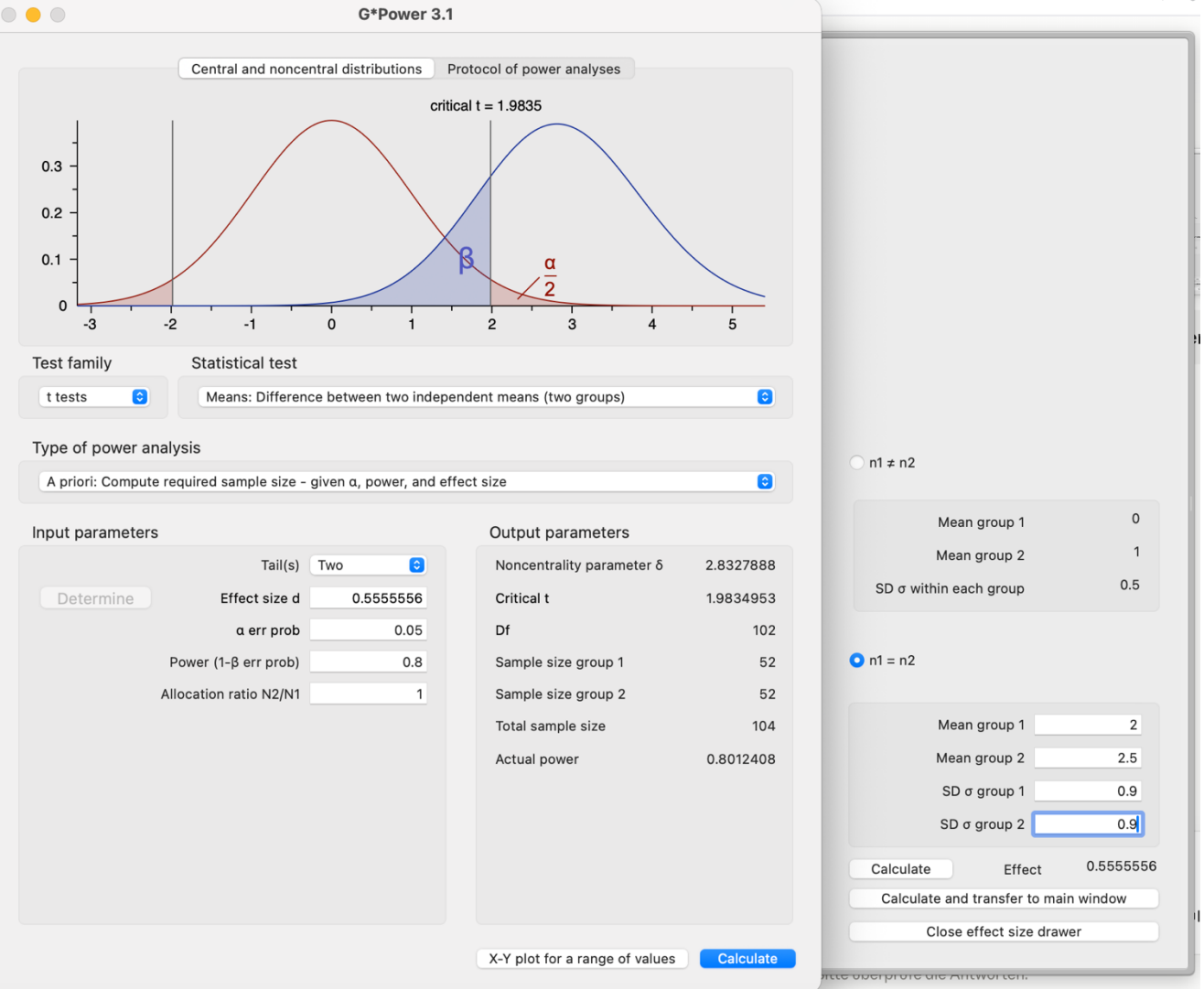
Click Determine, enter two means 0.5 mm apart (e.g. 3.5 and 4.0) and SD 0.9 for both groups: d = 0.556 and 52 per group – the power.t.test answer.

SAMPLE SIZE

Now it matches R

SD 0.9 in both fields: the drawer computes d = 0.5556 and transfers it.

52 per group, 104 in total – the power.t.test answer.



Click Calculate twice

Once in the drawer, once in the main window – the drawer never recalculates on its own.

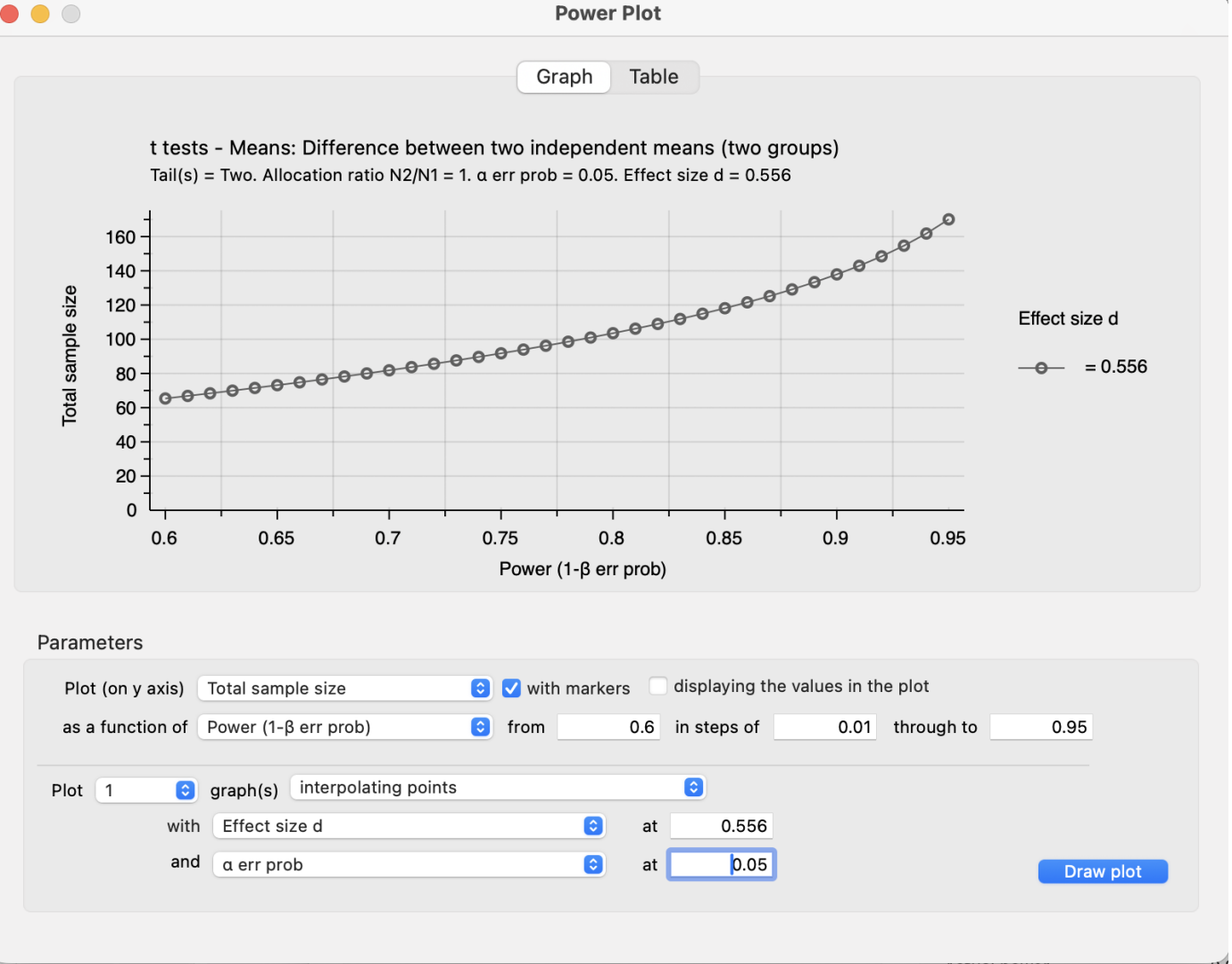
SAMPLE SIZE

The price of power

Same effect (d = 0.556), same alpha – only the power target moves:

0.80 → 104 patients
0.90 → 140 patients
0.95 → 172 patients

Read off the curve before you promise a power to the ethics committee.



What this formula does not cover

Randomised trial: covered

Randomisation removes confounding in expectation – the two-group formula is exactly right. Adjusting for baseline probing depth even gains power: it shrinks the residual SD.

Observational comparison: underpowered

Without randomisation the groups differ in more than the exposure: eg smokers are also older and brush less. Untangling the exposure from these differences costs patients – the more the groups differ, the more extra you need. **About a third more, or 10-15 patients per covariate**

Interaction: severely underpowered

An interaction asks whether the effect itself differs between subgroups – does the treatment work less well in smokers? Even a randomised trial is only sized for its main effect: the subgroup question splits it into four groups and compares two differences instead of two means – **at least 4× the patients, in practice often 16×.**

Powered for the crude comparison

This section sizes a two-group trial. An adjusted observational effect or a subgroup claim – even from a randomised trial – needs its own, larger calculation. Unplanned subgroup findings are exploratory.

Choosing the test: the one table to keep

Outcome	Groups	Test
Continuous - probing depth	Two, unpaired	Student's t-test
Continuous - split mouth	Two, paired	Paired t-test
Continuous	More than two, unpaired	ANOVA
Ordinal - plaque index, pain grade	Two, unpaired	Mann-Whitney U
Ordinal - before and after	Two, paired	Wilcoxon signed-rank
Ordinal	More than two, unpaired	Kruskal-Wallis
Ordinal	More than two, paired	Friedman test
Dichotomous - denture yes / no	Two, unpaired	Chi-square or Fisher
Dichotomous - split mouth	Two, paired	McNemar
Time to implant failure	Two or more	Kaplan-Meier and log-rank

HANDS-ON

Thirty minutes in Posit Cloud

	Exercise	What you run
1	The twenty patients with probing depth	mean, sd, median, IQR, boxplot, the CI by hand
2	Look at the shape, then compare the protocols	qqnorm, boxplot, t.test - unpaired and paired
3	When the t-test does not apply	wilcox.test, chisq.test, fisher.test
4	How many patients for the next study?	power.t.test, forwards and backwards
5	Do two measurements move together?	plot, then cor.test - Pearson and Spearman

day2-exercises.R

No data files and no packages to install

6

Ordinal data

Pain after third molar surgery: the Mann-Whitney U test

When the parametric route is closed

- Pain graded none / mild / moderate / severe: nobody can say that mild to moderate is the same step as moderate to severe.
- Small samples, where the shape of the distribution cannot be checked at all.
- Skewed measurements, where one patient drags the mean away from everyone else.

The alternative

Forget the values and use their order: the test asks whether one group tends to rank above the other, and the null distribution is built from the ranks themselves. It is not a test of the two medians.

How the ranking works

Pain grade	1	2	3	4	
Ranks it occupies	1 - 7	8 - 10	11 - 15	16 - 20	
Shared rank	4 (1+…+7) / 7	9 (8+9+10) / 3	13 (11+…+15) / 5	18 (16+…+20) / 5	
Ibuprofen patients	4	2	2	2	$n_1 = 10$
Paracetamol patients	3	1	3	3	$n_2 = 10$

Seven patients are tied at grade 1, so nothing distinguishes them: they take up rank positions 1 to 7, and each is given the average of those seven positions - not the first of them, and not the middle one. Every tied grade is handled the same way, so the twenty ranks still add up to 1 + 2 + … + 20 = 210, and so do $R_1 + R_2$.

$$R_1 = 4\times4 + 2\times9 + 2\times13 + 2\times18 = 96 \qquad R_2 = 3\times4 + 1\times9 + 3\times13 + 3\times18 = 114$$

From R to U

$U_1 = R_1 - n_1(n_1+1)/2 = 96 - 55 = 41$, and $U_2 = 114 - 55 = 59$. U_1 counts the pairs in which an ibuprofen patient had the higher grade, ties counting half, so $U_1 + U_2 = n_1 \times n_2 = 100$ and 50 each would be a dead heat. R calls it W.

Mann-Whitney in R

```
# pain at 24 hours, graded 1 (none) to 4 (severe)
pa <- c(1, 1, 2, 1, 2, 4, 4, 1, 3, 3)      # ibuprofen
pb <- c(1, 4, 2, 1, 3, 4, 4, 1, 3, 3)      # paracetamol

median(pa); median(pb)      # 2 and 3
table(pa); table(pb)        # report the whole distribution
wilcox.test(pa, pb)         # Mann-Whitney U, two-sided
```

What comes back

W = 41, p = 0.50: no evidence of a difference.

7

Counts and proportions

Dentures in the over-seventies: chi-square and Fisher's exact test

The 2 x 2 contingency table

Patients over 70	Complete denture	No denture	Total
Men	10	40	50
Women	38	32	70
Total	48	72	120

What the table says

20% of the men wear a complete denture against 54% of the women - a gap of 34 percentage points. Big enough to see by eye, but the question is still whether chance alone could produce it from 120 patients.

The same table if H0 were true

Expected counts	Complete denture	No denture	Total
Men	20 50 x 48 / 120	30 50 x 72 / 120	50
Women	28 70 x 48 / 120	42 70 x 72 / 120	70
Total	48	72	120

How the four cells are filled in

H0 says men and women are one population, so both rows must show the same proportion: 48 of 120, or 40%. Every cell is then row total x column total / grand total. Note what did not move - all four totals are identical to the observed table. Only the inside changed.

Observed against expected

chi-square = sum over the four cells of (observed - expected)² / expected

$(10 - 20)^2 / 20$	$(40 - 30)^2 / 30$	$(38 - 28)^2 / 28$	$(32 - 42)^2 / 42$
5.00	**3.33**	**3.57**	**2.38**

chi-square = 14.29 on 1 degree of freedom, p = 0.0002

Where the one degree of freedom comes from

The four totals are fixed by the study, so the cells are not free to vary independently. Write any one of them down - the 10 in the top left - and the other three follow by subtraction. One free cell, one degree of freedom. In general (rows - 1) x (columns - 1), so every 2 x 2 table has exactly one.

Chi-square and Fisher in R

```
tab <- matrix(c(10, 40, 38, 32), nrow = 2, byrow = TRUE)
dimnames(tab) <- list(sex = c('men','women'),
                      denture = c('yes','no'))
tab
prop.table(tab, 1) * 100            # 20% and 54%

chisq.test(tab, correct = FALSE)    # chi-square 14.29, p = 0.0002
```

When the counts are small

- Chi-square is an approximation, and it fails when the expected counts are small.
- Use Fisher’s exact test whenever an expected count falls below 5.
- The decision rests on the expected counts, not on the observed ones and not on the total sample size.

An example from a small practice study

Pulpal anaesthesia was achieved in 8 of 10 patients with articaine and 3 of 10 with mepivacaine. Expected counts 5.5 and 4.5: uncorrected chi-square gives p = 0.02, Yates’ 0.07, Fisher 0.07.

8

Time to an event

Short implants against long implants with sinus lift

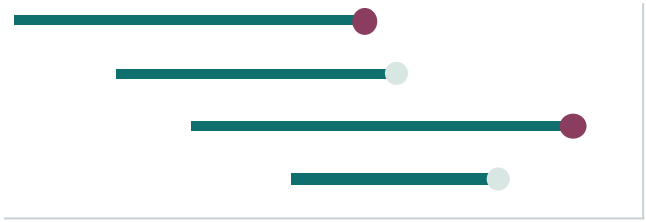
Why survival data are different

- The outcome is not whether an implant failed, but when.
- Patients enter the study at different times, and the study stops on a fixed date.
- Some move away, or simply stop attending, before anything has happened.

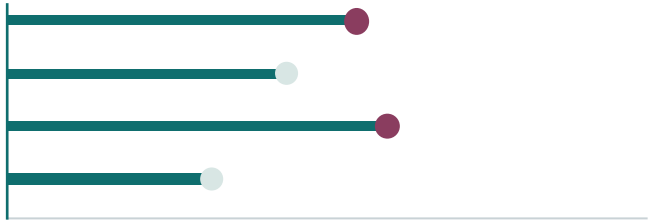
Censoring

A patient followed for two years without failure is neither a success nor a failure - the implant was still in function when we last looked. Dropping those patients, or counting them as failures, biases the answer in opposite directions.

Calendar time and serial time



as the trial ran: entry, and a closing date



as the analysis sees it: everyone starts at zero

- Patients are recruited over years and the study closes on a single date, so no two patients are followed for the same length of time.
- Every patient is moved back to their own time zero - the day the implant was placed - and only the duration of known survival counts.

Three variables per patient

The serial time, the status at the end of it (1 = failed, 0 = censored) and the group.

What each patient contributes

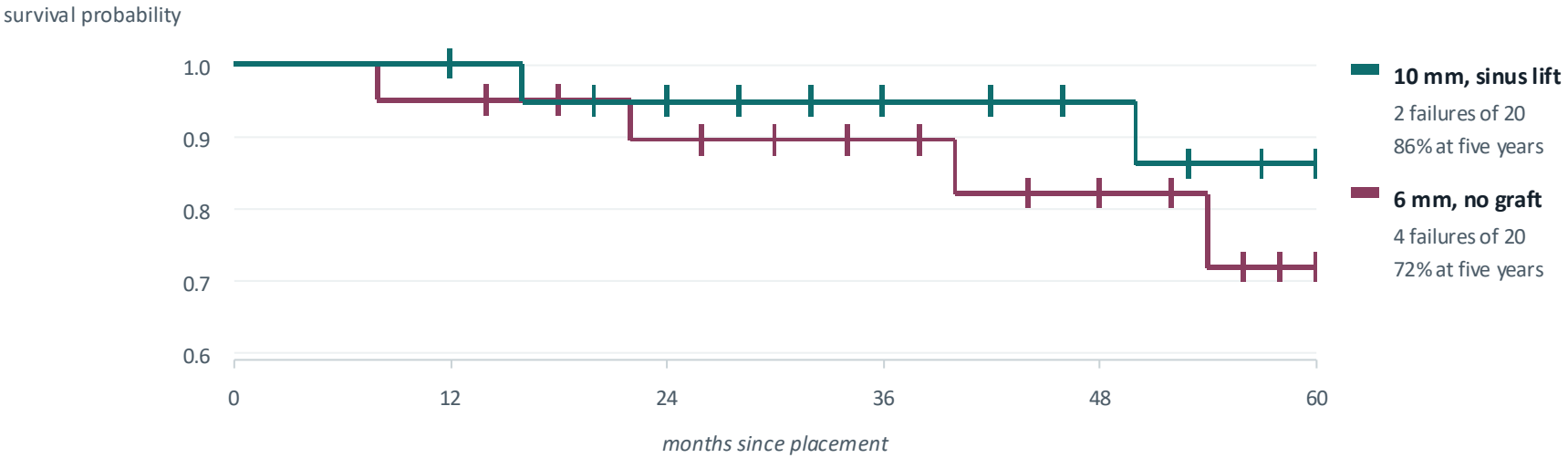
Implant	Months in function	Status	Length
1	8	failed	6 mm
2	14	censored - moved away	6 mm
3	40	failed	6 mm
4	60	censored - study closed	6 mm
5	16	failed	10 mm
6	42	censored - stopped attending	10 mm
7	60	censored - study closed	10 mm

Seven rows out of forty: the study had 20 implants of each length, one row each, followed for up to five years.

Censoring

An implant still in function when the study closed, or when the patient stopped attending, is neither a success nor a failure. It counts for as long as it was watched and then leaves the group at risk.

Two curves: 6 mm against 10 mm



Log-rank: p = 0.36

A visible gap, and still no difference shown in log-rank test

How the curve is built: the 6 mm group

Month	At risk	What happened	Survival
8	20	1 failure	19/20 = 0.950
14, 18		2 censored	0.950
22	17	1 failure	0.950 x 16/17 = 0.894
26 - 38		4 censored	0.894
40	12	1 failure	0.894 x 11/12 = 0.820
44 - 52		3 censored	0.820
54	8	1 failure	0.820 x 7/8 = 0.717

Why the steps get bigger

At each failure, multiply the survival so far by the proportion of those still at risk who got through. Censoring never makes the curve step down - but every censored implant leaves the group at risk, so the next failure is one out of a smaller group. One implant out of 20 costs 5.0 points; one out of 8 costs 10.2.

Kaplan-Meier in R

```
library(survival)                    # ships with R, just load it

# months in function; status: 1 = failed, 0 = still in function
fit <- survfit(Surv(months, status) ~ length, data = d)

plot(fit)                            # the two curves
summary(fit)                         # the table behind them
survdiff(Surv(months, status) ~ length, data = d)      # chi-square 0.83, p = 0.36
```

The log-rank test

Compares the whole curves, not survival at one chosen month

9

Correlation

Age and marginal bone loss: Pearson and Spearman

One number for how tightly points follow a line

r = +0.9

tight, and rising



r = 0

no linear pattern



r = −0.9

tight, and falling



- Each patient contributes two measurements - say age and marginal bone loss - and appears as one point.
- r runs from −1 to +1: the sign gives the direction, the size the tightness - and r = 0 means no straight-line pattern, not no pattern at all.

Pearson or Spearman

Pearson's r

Continuous measurements, roughly normal, and a relationship that is a straight line. Probing depth against radiographic bone loss

Spearman's r_s

Works on the ranks, so it asks only whether more of one goes with more of the other. For ordinal data - plaque index, pain grades - and for skewed measurements with outliers.

In R

cor(x, y) gives the number; cor.test(x, y) adds the 95% CI and the p-value; method = "spearman" switches to ranks. The p answers only whether r is zero - the interval says how well r is pinned down.

What r does not tell you

- Correlation is not causation: coffee and heart disease move together because smoking drives both - the confounder from day 1.
- r sees straight lines only: a U-shape scores near zero, and one outlying patient can manufacture - or destroy - an impressive r in a small sample.
- r^2 is the share of the variance explained: r = 0.7 sounds strong, yet half of the variation is still unexplained.

Correlation is not agreement

Two examiners can correlate at 0.95 and still differ by a full millimetre on every pocket. **Agreement is a different question**: **Bland-Altman for measurements, kappa for categories**.

Four things to take away

1. **The scale of the data decides everything**
   Nominal, ordinal, interval or ratio - it fixes the summary, the figure and the test before you start.

2. **Describe before you test**
   Mean and median far apart, or two peaks in the histogram, matter more than any p-value.

3. **Alpha is a false alarm, beta is a missed difference**
   Power is 1 - beta, and it is bought with patients before the study, not after.

4. **Report the effect with its confidence interval**
   In millimetres, with the interval beside it - a p-value alone tells the reader neither how large the difference is nor how precisely you know it.